

Experiment 5

AIM: Deploy a simple application using Nginx / Apache.

```
C:\Users\manis>mkdir nginx-demo
```

```
C:\Users\manis>cd nginx-demo
```

```
C:\Users\manis\nginx-demo>npm init -y
```

Wrote to C:\Users\manis\nginx-demo\package.json:

```
{
  "name": "nginx-demo",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

```
C:\Users\manis\nginx-demo>npm i express
```

added 65 packages, and audited 66 packages in 5s

22 packages are looking for funding
run `npm fund` for details

found 0 vulnerabilities

```
C:\Users\manis\nginx-demo>node index.js
```

App running on <http://localhost:3000>

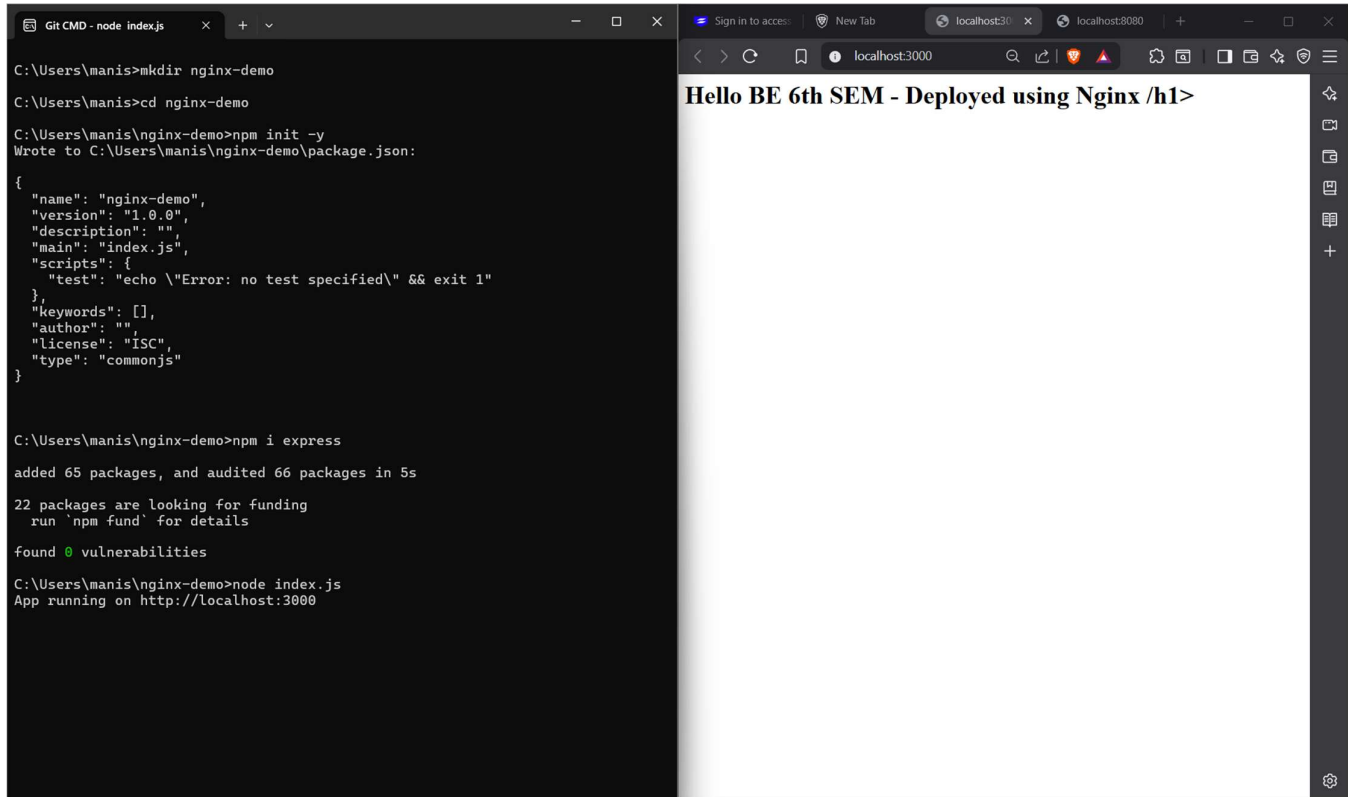
Open Another CMD

```
C:\nginx-1.29.5>nginx -v
nginx version: nginx/1.29.5
```

```
C:\nginx-1.29.5>cd conf
```

```
C:\nginx-1.29.5\conf>notepad nginx.conf
```

```
C:\nginx-1.29.5\conf>
```



The screenshot shows a development environment with two windows. The left window is a terminal running Git CMD, and the right window is a web browser.

Terminal Output:

```
C:\Users\manis>mkdir nginx-demo
C:\Users\manis>cd nginx-demo
C:\Users\manis\nginx-demo>npm init -y
Wrote to C:\Users\manis\nginx-demo\package.json:

{
  "name": "nginx-demo",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

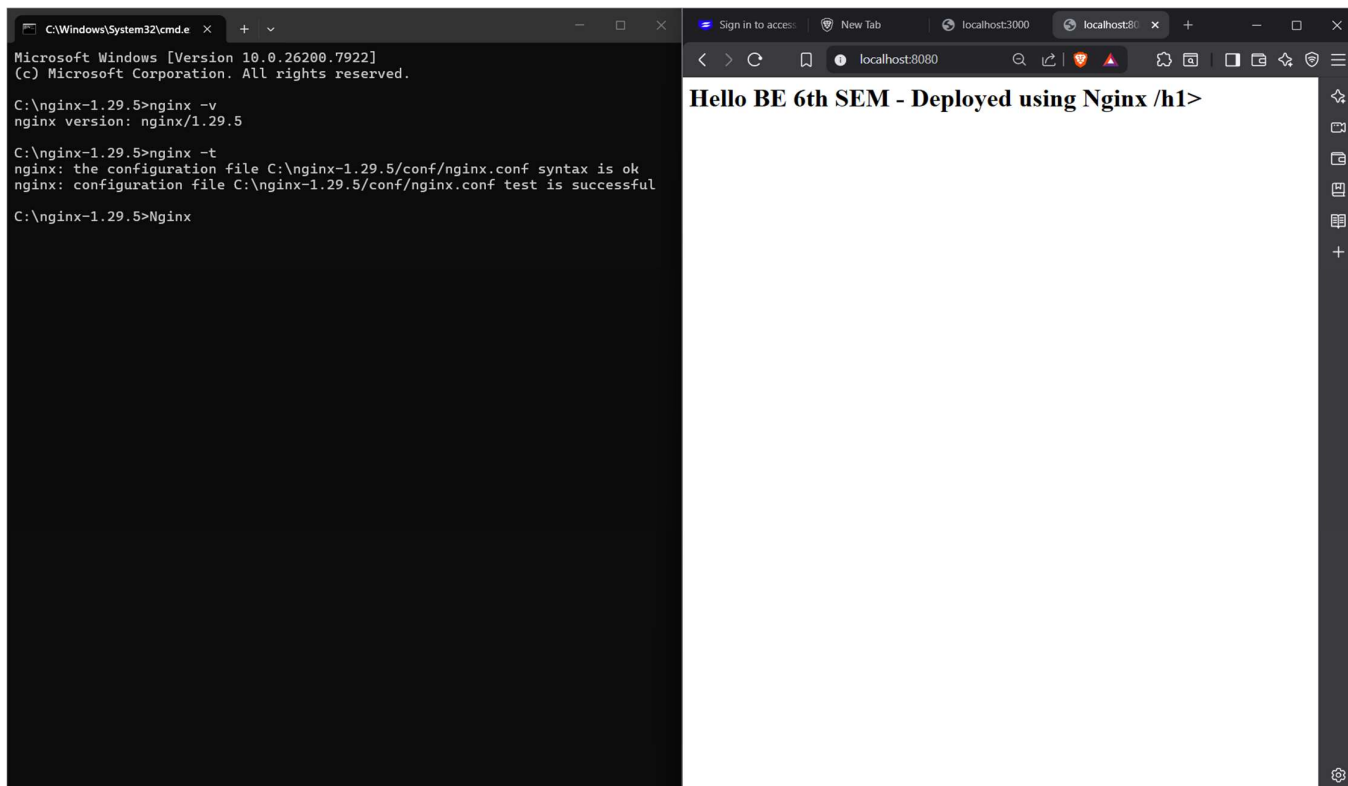
C:\Users\manis\nginx-demo>npm i express
added 65 packages, and audited 66 packages in 5s
22 packages are looking for funding
  run 'npm fund' for details
found 0 vulnerabilities

C:\Users\manis\nginx-demo>node index.js
App running on http://localhost:3000
```

Browser Output:

localhost:3000

Hello BE 6th SEM - Deployed using Nginx /h1>



The screenshot shows a development environment with two windows. The left window is a terminal running Windows CMD, and the right window is a web browser.

Terminal Output:

```
C:\Windows\System32\cmd.e x + v
Microsoft Windows [Version 10.0.26200.7922]
(c) Microsoft Corporation. All rights reserved.

C:\nginx-1.29.5>nginx -v
nginx version: nginx/1.29.5

C:\nginx-1.29.5>nginx -t
nginx: the configuration file C:\nginx-1.29.5/conf/nginx.conf syntax is ok
nginx: configuration file C:\nginx-1.29.5/conf/nginx.conf test is successful

C:\nginx-1.29.5>Nginx
```

Browser Output:

localhost:8080

Hello BE 6th SEM - Deployed using Nginx /h1>

Experiment 6

AIM: Setup Nginx to handle proxy requests and load balancing

```
C:\Users\manis\load-balance-demo>npm i express  
added 65 packages, and audited 66 packages in 2s  
  
22 packages are looking for funding  
  run `npm fund` for details  
  
found 0 vulnerabilities
```

Server-1

```
C:\Users\manis\load-balance-demo>mkdir server1  
  
C:\Users\manis\load-balance-demo>cd server1  
  
C:\Users\manis\load-balance-demo\server1>npm init -y  
Wrote to C:\Users\manis\load-balance-demo\server1\package.json:  
  
{  
  "name": "server1",  
  "version": "1.0.0",  
  "description": "",  
  "main": "index.js",  
  "scripts": {  
    "test": "echo \"Error: no test specified\" && exit 1"  
  },  
  "keywords": [],  
  "author": "",  
  "license": "ISC",  
  "type": "commonjs"  
}
```

```
C:\Users\manis\load-balance-demo\server1>npm i express  
added 65 packages, and audited 66 packages in 1s  
  
22 packages are looking for funding  
  run `npm fund` for details  
  
found 0 vulnerabilities  
  
C:\Users\manis\load-balance-demo\server1>notepad server.js  
  
C:\Users\manis\load-balance-demo\server1>node server.js  
Server running on port 3000
```

Server-2

```
C:\Users\manis>cd load-balance-demo

C:\Users\manis\load-balance-demo>mkdir server2

C:\Users\manis\load-balance-demo>cd server2

C:\Users\manis\load-balance-demo\server2>npm init -y
Wrote to C:\Users\manis\load-balance-demo\server2\package.json:

{
  "name": "server2",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

C:\Users\manis\load-balance-demo\server2>npm i express

added 65 packages, and audited 66 packages in 3s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Users\manis\load-balance-demo\server2>notepad server.js

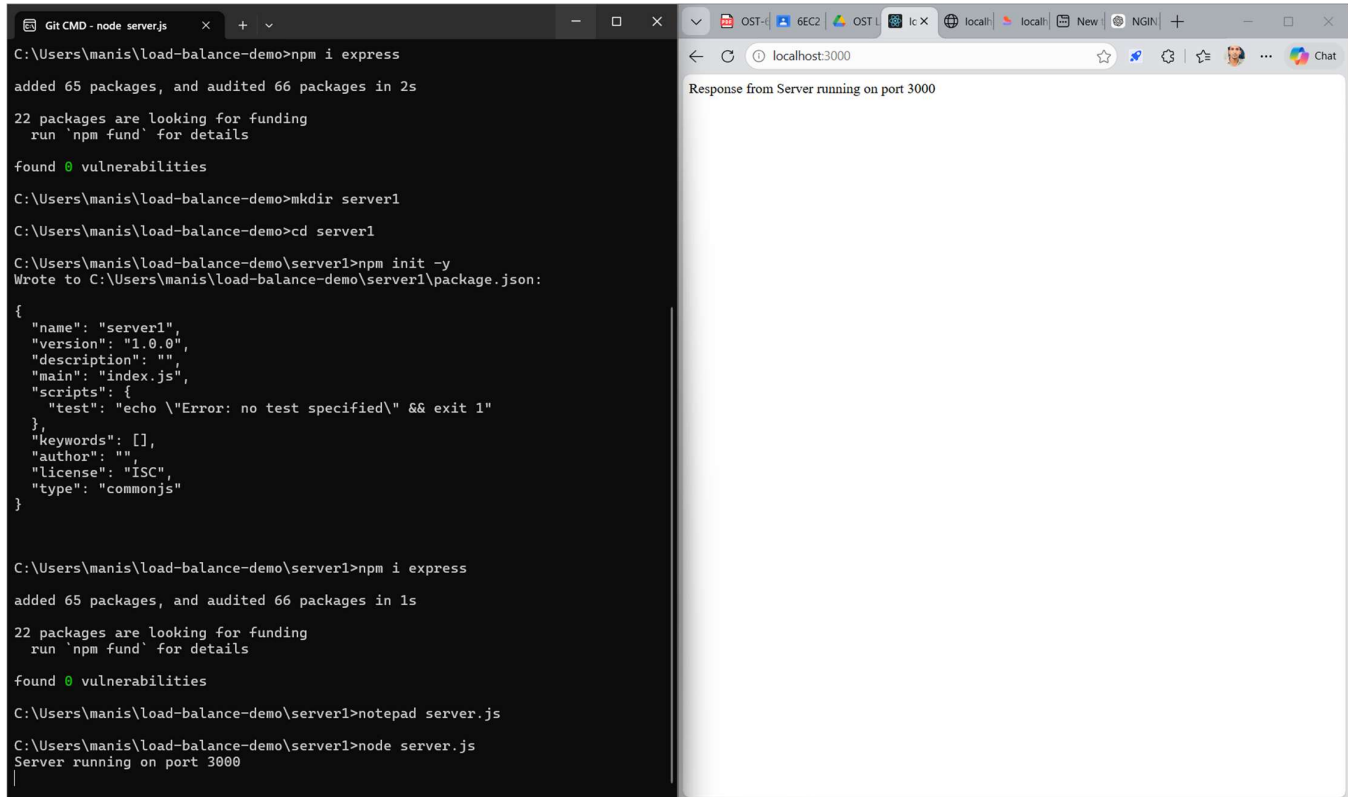
C:\Users\manis\load-balance-demo\server2>node server.js
Server running on port 3001
```

Nginx

```
Microsoft Windows [Version 10.0.26200.7922]
(c) Microsoft Corporation. All rights reserved.

C:\nginx-1.29.5>nginx -s stop

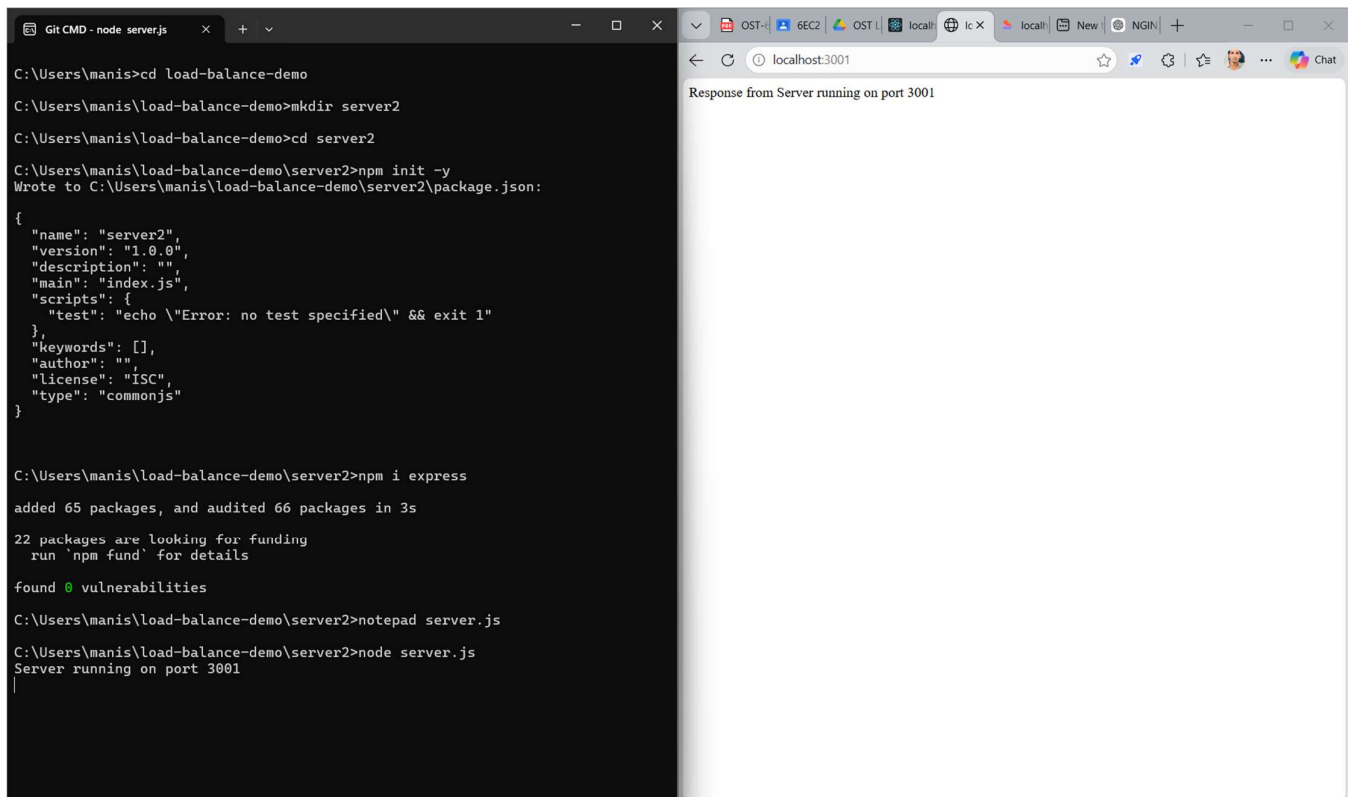
C:\nginx-1.29.5>nginx
```



The screenshot displays a Windows command prompt and a web browser. In the command prompt, the user navigates to the 'load-balance-demo' directory, creates a 'server1' subdirectory, and enters it. They then initialize a new Node.js project with 'npm init -y', which generates a 'package.json' file. The file content is shown as follows:

```
{
  "name": "server1",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \\\"Error: no test specified\\\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

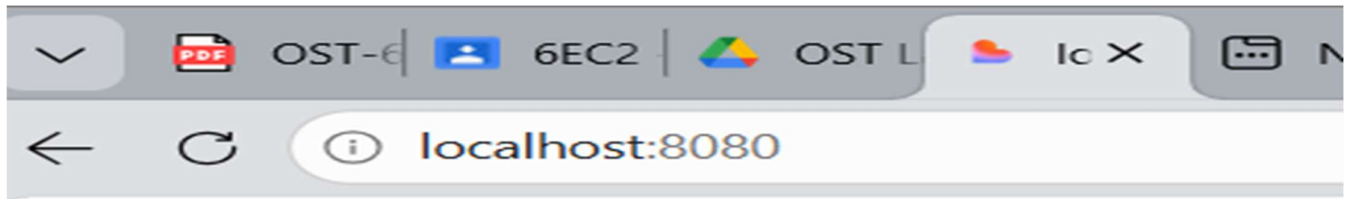
Next, the user installs Express.js with 'npm i express'. Finally, they create a 'server.js' file using 'notepad server.js' and run it with 'node server.js', resulting in the message 'Server running on port 3000'. The web browser on the right shows the address 'localhost:3000' and the response 'Response from Server running on port 3000'.



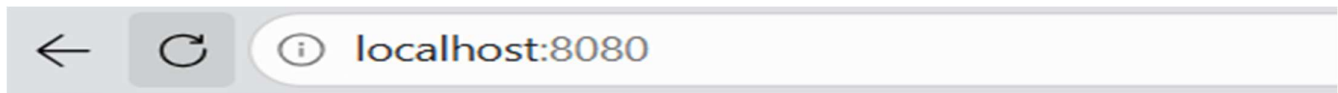
The screenshot shows the same development environment as the previous one, but for a second server. The user navigates to 'load-balance-demo', creates a 'server2' subdirectory, and enters it. They initialize a new Node.js project with 'npm init -y', generating a 'package.json' file with the following content:

```
{
  "name": "server2",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \\\"Error: no test specified\\\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

The user then installs Express.js with 'npm i express'. They create a 'server.js' file using 'notepad server.js' and run it with 'node server.js', resulting in the message 'Server running on port 3001'. The web browser on the right shows the address 'localhost:3001' and the response 'Response from Server running on port 3001'.



Slow Server - running on port 3000



Response from Fast Server running on port 3001

Experiment 7

AIM: Use Selenium to create browser-based tests.

```
C:\Users\manis\login-testing>pip install selenium
Collecting selenium
  Downloading selenium-4.41.0-py3-none-any.whl.metadata (7.5 kB)
Collecting certifi>=2026.1.4 (from selenium)
  Downloading certifi-2026.2.25-py3-none-any.whl.metadata (2.5 kB)
Collecting trio<1.0,>=0.31.0 (from selenium)
  Downloading trio-0.33.0-py3-none-any.whl.metadata (8.5 kB)
Collecting trio-websocket<1.0,>=0.12.2 (from selenium)
  Downloading trio_websocket-0.12.2-py3-none-any.whl.metadata (5.1 kB)
Collecting typing_extensions<5.0,>=4.15.0 (from selenium)
  Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Collecting urllib3<3.0,>=2.6.3 (from urllib3[socks]<3.0,>=2.6.3->selenium)
  Using cached urllib3-2.6.3-py3-none-any.whl.metadata (6.9 kB)
Requirement already satisfied: websocket-client<2.0,>=1.8.0 in
C:\Users\manis\AppData\Local\Programs\Python\Python312\Lib\site-packages (from
selenium) (1.8.0)
Requirement already satisfied: attrs>=23.2.0 in
C:\Users\manis\AppData\Local\Programs\Python\Python312\Lib\site-packages (from
trio<1.0,>=0.31.0->selenium) (24.1.0)
Collecting sortedcontainers (from trio<1.0,>=0.31.0->selenium)
  Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: idna in
C:\Users\manis\AppData\Local\Programs\Python\Python312\Lib\site-packages (from
trio<1.0,>=0.31.0->selenium) (3.7)
Collecting outcome (from trio<1.0,>=0.31.0->selenium)
  Downloading outcome-1.3.0.post0-py2.py3-none-any.whl.metadata (2.6 kB)
Requirement already satisfied: sniffio>=1.3.0 in
C:\Users\manis\AppData\Local\Programs\Python\Python312\Lib\site-packages (from
trio<1.0,>=0.31.0->selenium) (1.3.1)
Requirement already satisfied: cffi>=1.14 in
C:\Users\manis\AppData\Local\Programs\Python\Python312\Lib\site-packages (from
trio<1.0,>=0.31.0->selenium) (1.16.0)
Collecting wsproto>=0.14 (from trio-websocket<1.0,>=0.12.2->selenium)
  Downloading wsproto-1.3.2-py3-none-any.whl.metadata (5.2 kB)
Collecting pysocks!=1.5.7,<2.0,>=1.5.6 (from urllib3[socks]<3.0,>=2.6.3-
>selenium)
  Downloading PySocks-1.7.1-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: pycparser in
C:\Users\manis\AppData\Local\Programs\Python\Python312\Lib\site-packages (from
cffi>=1.14->trio<1.0,>=0.31.0->selenium) (2.22)
Collecting h11<1,>=0.16.0 (from wsproto>=0.14->trio-websocket<1.0,>=0.12.2-
>selenium)
  Using cached h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
Downloading selenium-4.41.0-py3-none-any.whl (9.5 MB)
9.5/9.5 MB 827.5 kB/s 0:00:11
Downloading trio-0.33.0-py3-none-any.whl (510 kB)
Downloading trio_websocket-0.12.2-py3-none-any.whl (21 kB)
Using cached typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Using cached urllib3-2.6.3-py3-none-any.whl (131 kB)
Downloading PySocks-1.7.1-py3-none-any.whl (16 kB)
Downloading certifi-2026.2.25-py3-none-any.whl (153 kB)
Downloading outcome-1.3.0.post0-py2.py3-none-any.whl (10 kB)
```

```
Downloading wsproto-1.3.2-py3-none-any.whl (24 kB)
Using cached h11-0.16.0-py3-none-any.whl (37 kB)
Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl (29 kB)
Installing collected packages: sortedcontainers, urllib3, typing_extensions,
pysocks, outcome, h11, certifi, wsproto, trio, trio-websocket, selenium
  Attempting uninstall: urllib3
    Found existing installation: urllib3 2.2.2
    Uninstalling urllib3-2.2.2:
      Successfully uninstalled urllib3-2.2.2
  Attempting uninstall: h11
    Found existing installation: h11 0.14.0
    Uninstalling h11-0.14.0:
      Successfully uninstalled h11-0.14.0
  Attempting uninstall: certifi
    Found existing installation: certifi 2024.7.4
    Uninstalling certifi-2024.7.4:
      Successfully uninstalled certifi-2024.7.4
ERROR: pip's dependency resolver does not currently take into account all the
packages that are installed. This behaviour is the source of the following
dependency conflicts.
httpcore 1.0.5 requires h11<0.15,>=0.13, but you have h11 0.16.0 which is
incompatible.
Successfully installed certifi-2026.2.25 h11-0.16.0 outcome-1.3.0.post0
pysocks-1.7.1 selenium-4.41.0 sortedcontainers-2.4.0 trio-0.33.0 trio-
websocket-0.12.2 typing_extensions-4.15.0 urllib3-2.6.3 wsproto-1.3.2
```

```
C:\Users\manis\login-testing>pip install webdriver-manager
Collecting webdriver-manager
  Downloading webdriver_manager-4.0.2-py2.py3-none-any.whl.metadata (12 kB)
Requirement already satisfied: requests in
C:\Users\manis\AppData\Local\Programs\Python\Python312\Lib\site-packages (from
webdriver-manager) (2.32.3)
Collecting python-dotenv (from webdriver-manager)
  Downloading python_dotenv-1.2.2-py3-none-any.whl.metadata (27 kB)
Requirement already satisfied: packaging in
C:\Users\manis\AppData\Local\Programs\Python\Python312\Lib\site-packages (from
webdriver-manager) (24.1)
Requirement already satisfied: charset-normalizer<4,>=2 in
C:\Users\manis\AppData\Local\Programs\Python\Python312\Lib\site-packages (from
requests->webdriver-manager) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in
C:\Users\manis\AppData\Local\Programs\Python\Python312\Lib\site-packages (from
requests->webdriver-manager) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in
C:\Users\manis\AppData\Local\Programs\Python\Python312\Lib\site-packages (from
requests->webdriver-manager) (2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in
C:\Users\manis\AppData\Local\Programs\Python\Python312\Lib\site-packages (from
requests->webdriver-manager) (2026.2.25)
Downloading webdriver_manager-4.0.2-py2.py3-none-any.whl (27 kB)
Downloading python_dotenv-1.2.2-py3-none-any.whl (22 kB)
Installing collected packages: python-dotenv, webdriver-manager
Successfully installed python-dotenv-1.2.2 webdriver-manager-4.0.2
```

```
C:\Users\manis\login-testing>notepad salenium_login_test.py
```

```
C:\Users\manis\login-testing>python selenium_login_test.py
python: can't open file 'C:\\Users\\manis\\login-
testing\\selenium_login_test.py': [Errno 2] No such file or directory
```

```
C:\Users\manis\login-testing>python salenium_login_test.py
Login Message: You logged into a secure area!
x
```

```
C:\Users\manis\login-testing>
```

```
C:\Users\manis\login-testing>python salenium_login_test.py
Login Message: You logged into a secure area!
x
```